1. Student Details

Team Details	Information
Batch Number	3
Names	Preethaa Vijayakumar Reshmitha Y Voonadi Sree Raghavi Vruti Thakkar
Internship Duration	8 Weeks
Project	Supply Chain Visibility System With Optimization Analytics

2. Introduction

The project focuses on developing a Supply Chain Visibility System with Optimization Analytics using Microsoft Power BI. It provides end-to-end visibility into inventory, suppliers, warehouses, transportation, shipments, and delivery operations.

The project is needed to bring information from multiple supply chain tables into an interactive analytical dashboard. It helps users understand operational performance through KPIs, charts, filters, and drill-down analysis.

The dashboard addresses the problem of analyzing large and distributed supply chain data by presenting important measures such as revenue, orders, inventory, delivery performance, shipping cost, supplier reliability, transportation cost, and fulfillment.

The internship work is organized into milestones covering data modelling and KPI foundation, inventory and delivery analytics, and supplier and transportation analytics. The available report content is reorganized here up to Milestone 3.

3. Tools Used

- Power BI
- Power Query
- DAX
- Microsoft Excel
- Data Visualization
- Data Modelling

4. Milestone 1 – Data Modelling & KPI Foundation

4.1 Milestone 1 Introduction

The first milestone focused on understanding the Supply Chain Management dataset and developing an initial supply chain performance dashboard using Microsoft Power BI. The work included data understanding, data cleaning and preprocessing, data modelling, creation of relationships, development of important KPIs and DAX measures, and visualization of the results.

The dashboard provides an initial overview of supply chain performance through revenue, orders, customers, products, delivery time, supplier reliability, shipping cost, delivery status, state-wise orders, and category-wise revenue.

Overall flow: Supply Chain Data → Data Cleaning → Data Modelling → KPI Development → Power BI Dashboard → Initial Business Insights.

4.2 Data Cleaning

Data cleaning was performed before creating the dashboard to improve data quality and ensure that KPIs, DAX measures, and visualizations were based on appropriate data.

- Checked missing, null, duplicate, incorrect, and inconsistent values.
- Checked and handled invalid values, including negative delivery/lead-time values.
- Cleaned date columns and converted them into proper date format.
- Corrected data types such as dates, whole numbers, and decimal values.
- Cleaned price and numerical columns.
- Verified product IDs and other key columns.
- Prepared the data for DAX measures and Power BI visualizations.

After preprocessing, the dataset became more consistent, structured, and suitable for analysis and visualization in Power BI.

4.3 Data Modelling

A structured data model was created in Power BI using Fact and Dimension tables. The model follows a star-schema-based structure, where fact tables are connected with relevant dimension tables through common key fields.

Table Type	Tables
Dimension Tables	dim_customers, dim_products, dim_suppliers, dim_warehouses
Fact Tables	fact_orders, fact_order_items, fact_shipments

Important relationships:

- Customers → Orders using customer_id
- Orders → Order Items using order_id
- Products → Order Items using product_id
- Warehouses → Orders using warehouse_id
- Orders → Shipments using order_id
- Suppliers → Products using supplier_id

These relationships allow Power BI to combine information from different tables, filter data across related tables, calculate KPIs accurately, and create charts using fields from different tables.

4.4 KPI Creation

Key Performance Indicators were created to provide a quick overview of supply chain performance.

KPI / Measure	Purpose
Total Revenue	Measures overall revenue generated.
Total Orders	Shows the number of orders.
Total Customers	Shows the number of customers.
Total Products	Shows the number of products.
Average Delivery Time	Measures delivery performance.
Supplier Reliability	Indicates supplier performance.
Shipping Cost	Measures overall shipping expenditure.

4.5 Milestone 1 Dashboard Screenshots

The Milestone 1 dashboard provides an interactive overview of supply chain performance using KPI cards and visualizations.

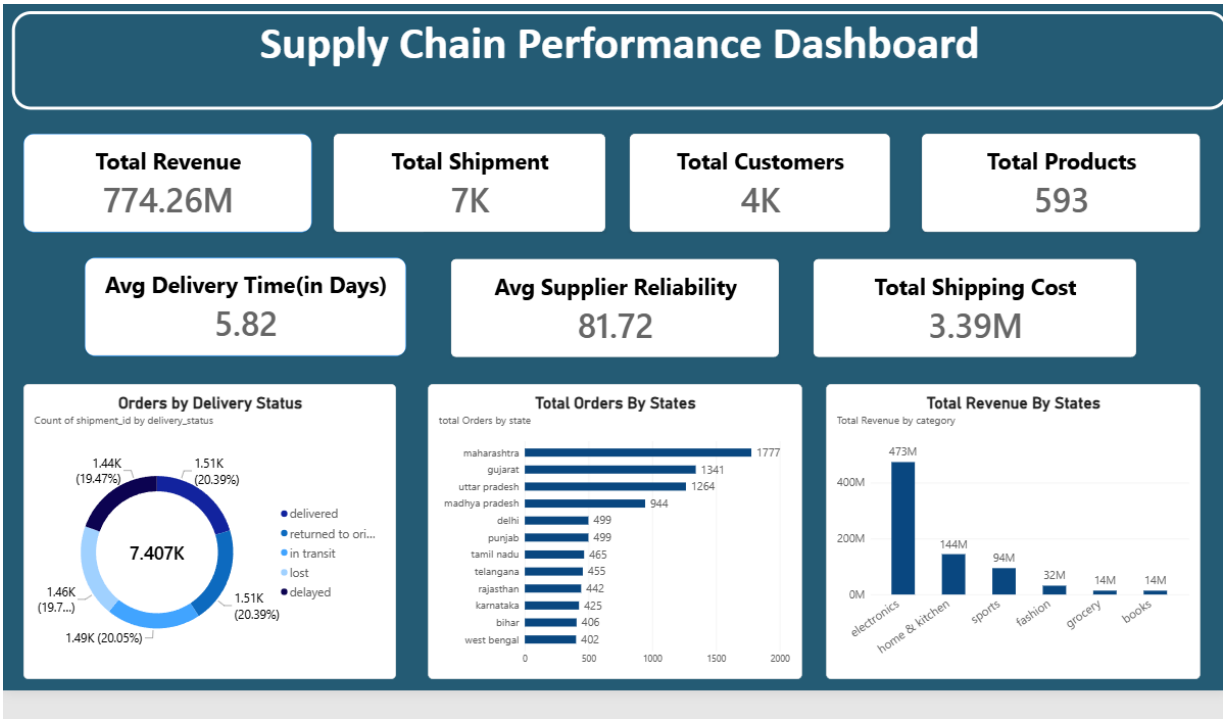


Figure 1: Milestone 1 – Supply Chain Performance Dashboard

The dashboard includes KPI cards for revenue, shipments, customers, products, average delivery time, supplier reliability, and shipping cost. It also presents delivery status, state-wise orders, and category-wise revenue.

5. Milestone 2 – Inventory & Delivery Analytics

5.1 Milestone 2 Introduction

Milestone 2 extended the Supply Chain Management analysis by focusing on inventory management and delivery performance. Separate Inventory Analysis and Delivery Analysis dashboards were developed in Power BI.

The Inventory Analysis Dashboard provides a visual view of inventory quantities, products, inventory value, inventory turnover, and slow-moving products. The Delivery Analysis Dashboard focuses on shipments, shipping cost, delivery status, delivered and delayed shipments, location-wise delivery performance, and shipment trends.

This milestone used the structured data model created earlier and added the required inventory quantity information to the product table for inventory analysis.

5.2 Inventory Analysis

Inventory data was not directly available in the original dataset in the required form. An Inventory_Quantity column was added to the product table as required for inventory analysis. The inventory quantity was then used with product unit price to calculate inventory value.

- Created the Inventory_Quantity column in the product table.
- Verified product IDs and inventory-related fields.
- Prepared inventory data for DAX calculations.
- Used product quantity and unit price for inventory value analysis.

Inventory KPIs and calculations:

Measure	Calculation / Purpose
Total Inventory	SUM(dim_products[Inventory_Quantity])
Total Products	DISTINCTCOUNT(dim_products[product_id])
Total Quantity Sold	SUM(fact_order_items[quantity])
Inventory Value	SUMX(dim_products, Inventory_Quantity × unit_price)
Inventory Turnover	DIVIDE([Total Revenue], [Inventory Value], 0)
Slow-Moving Products	Products with Total Quantity Sold <= 50

The Inventory Analysis Dashboard contains KPI cards for Total Inventory, Inventory Turnover, Total Products, and Slow-Moving Products. Visualizations include Inventory by Category, Top 5 Products by Inventory, and Inventory Trend Over Time.

Inventory analysis helps identify products with high stock levels, compare categories, understand inventory turnover, identify slow-moving products, and observe inventory trends.

5.3 Delivery Analysis

The delivery analysis used shipment information such as shipment ID, delivery status, shipping cost, delivery days, state, city, warehouse, and time-related fields. These fields were used to create delivery KPIs and visualizations.

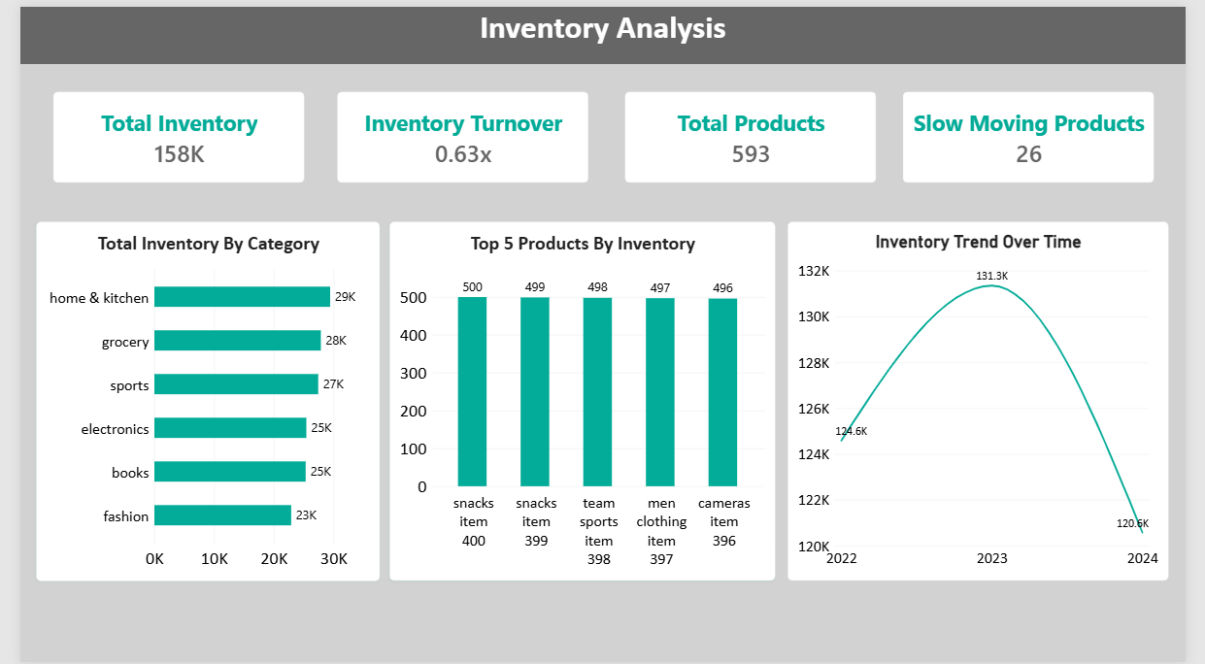
Measure	Calculation / Purpose
Total Shipments	COUNT(fact_shipments[shipment_id])
Total Shipping Cost	SUM(fact_shipments[shipping_cost])
Average Delivery Days	AVERAGE(fact_shipments[delivery_days])
Delivered Shipments	Shows the number of successfully delivered shipments.
Delayed Shipments	Shows the number of delayed shipments.

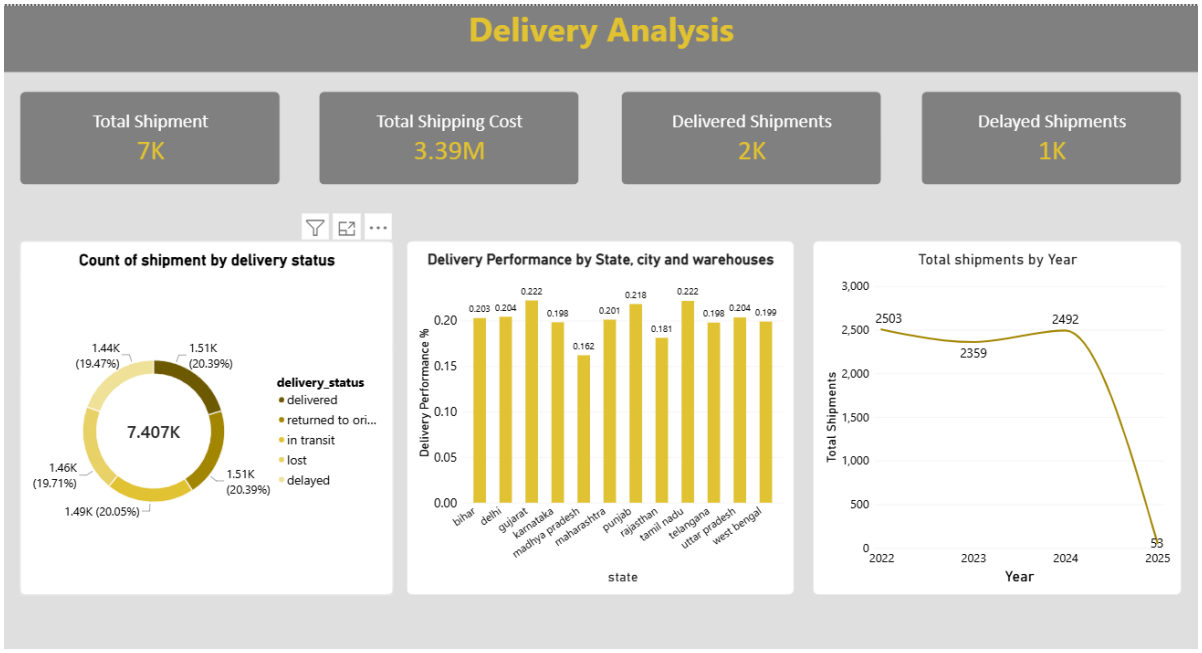
The Delivery Analysis Dashboard includes KPI cards for Total Shipping Cost, Total Shipments, Delivered Shipments, and Delayed Shipments. Visualizations include Shipment by Delivery Status, Delivery Performance by State, City and Warehouse, and Shipment Trend Over Time.

Delivery analysis helps identify delivered and delayed shipments, understand shipping expenditure, compare delivery performance across locations, and observe shipment activity over time.

5.4 Milestone 2 Dashboard Explanation

The Milestone 2 dashboards provide separate views for inventory and delivery analysis. The inventory dashboard focuses on stock quantity and value, while the delivery dashboard focuses on shipment status, cost, location-wise performance, and shipment trends.





6. Milestone 3 – Supplier & Transportation Analytics

6.1 Supplier Analysis

Milestone 3 includes supplier-focused analysis within the Shipments and Transportation Analysis Dashboard. Supplier information is used to compare shipment volumes and supplier reliability.

- Supplier-wise shipment analysis helps compare shipment volumes.
- Supplier reliability analysis provides information about supplier performance.
- A Top 5 Supplier visualization was created to compare suppliers.
- A Supplier slicer was added so the dashboard can be filtered for individual suppliers.

The dashboard also includes Average Supplier Reliability as a KPI/measure for supplier performance analysis.

6.2 Transportation Analysis

Milestone 3 focuses on analyzing shipments and transportation performance using Microsoft Power BI. The dashboard provides an overview of shipping costs, on-time shipments, order fulfillment, supplier performance, carrier performance, and country-wise transportation costs.

Interactive filters for Supplier, Carrier, and Country are used to analyze transportation performance from different perspectives.

- Carrier-wise analysis helps compare transportation costs.
- Country-wise shipment cost analysis helps compare transportation costs across countries.
- Shipping cost analysis provides an overview of transportation expenditure.
- Fulfillment and on-time shipment measures provide delivery-related transportation performance information.

6.3 KPI & DAX Measures

The following KPI measures were created for the Shipments and Transportation Analysis Dashboard:

KPI / Measure	Purpose
Total Shipping Cost	Measures the overall transportation expenditure.
Total Shipments	Shows the total number of shipments.
Average Shipping Cost	Shows the average shipping cost per shipment.
On Time Shipments	Counts shipments completed within the defined on-time delivery period.
On Time Rate	Measures the proportion of on-time shipments.
Total Orders	Shows the total number of orders.
Fulfillment Rate	Measures the proportion of fulfilled/delivered orders.
Average Supplier Reliability	Shows the average reliability score of suppliers.

Main DAX Measures

Total Shipping Cost

```
Total Shipping Cost =
SUM(fact_shipments[shipping_cost])
```

Total Shipments

```
Total Shipments =
COUNT(fact_shipments[shipment_id])
```

Average Shipping Cost

```
Avg Shipping Cost =
AVERAGE(fact_shipments[shipping_cost])
```

Total Orders

```
Total Orders =
COUNT(fact_orders[order_id])
```

On Time Shipments

```
On Time Shipments =
CALCULATE(
    COUNT(fact_shipments[shipment_id]),
    fact_shipments[delivery_days] <= 8
)
```

On Time Rate

```
On Time Rate =
DIVIDE(
    [On Time Shipments],
    [Total Shipments],
    0
)
```

Dashboard KPI values recorded in the report include Total Shipping Cost – 3.39M, On Time Rate – 0.83, On Time Shipments – 1K, Average Shipping Cost – 467.02, Total Orders – 9K, and Fulfillment Rate – 0.20.

Dashboard filters: Supplier, Carrier, Country.

6.4 Milestone 3 Dashboard Screenshots

The Shipments and Transportation Analysis Dashboard provides a visual overview of transportation and shipment performance, with supplier, carrier, and country filters.

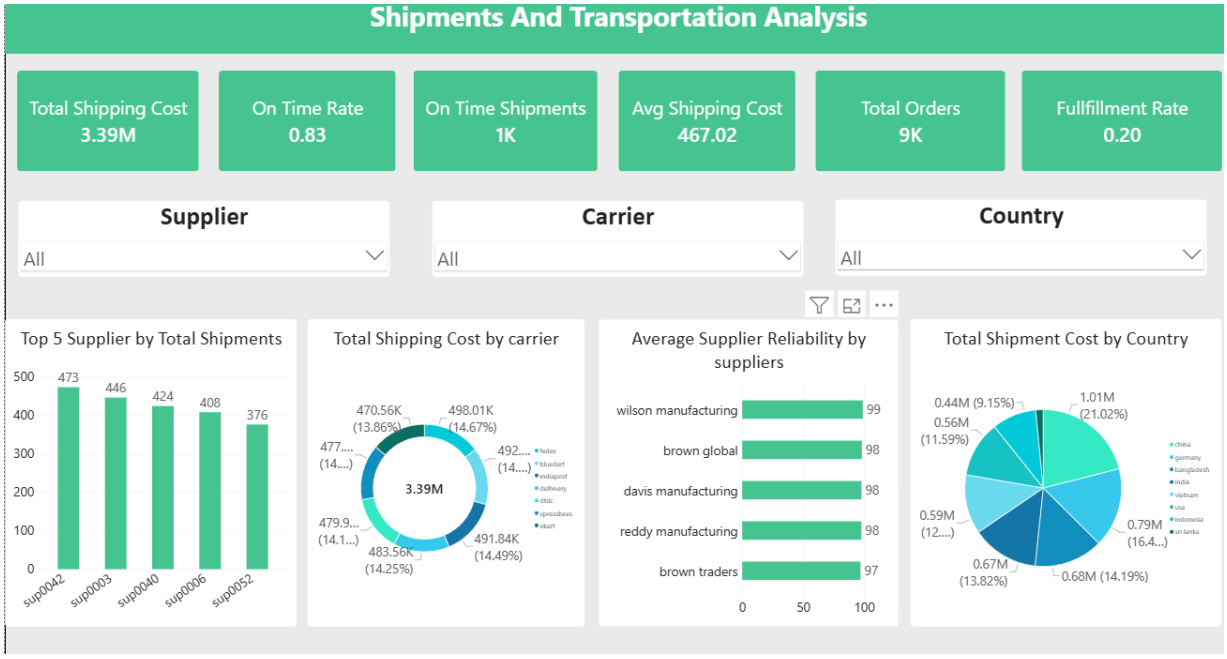


Figure 4: Milestone 3 – Shipments and Transportation Analysis Dashboard

The dashboard contains KPI cards for shipping cost, on-time rate, on-time shipments, average shipping cost, total orders, and fulfillment rate. It also includes supplier, carrier, and country analysis visuals.

7. Milestone 4 – Warehouse Efficiency, Executive Overview & Optimization Analytics

Milestone 4 focuses on analyzing overall supply chain performance using Microsoft Power BI. It includes warehouse efficiency analysis, an executive overview of key business indicators, and optimization analytics to support data-driven decision-making.

7.1 Warehouse Efficiency Analysis

The Warehouse Efficiency Dashboard focuses on comparing warehouse performance using order volumes, shipped quantities, on-time deliveries, and warehouse capacity.

- Warehouse-wise order analysis helps compare the number of orders handled by each warehouse.
- Warehouse capacity analysis provides an overview of the capacity assigned to different warehouses.
- On-time delivery analysis helps compare delivery performance across warehouses.
- Category-wise order analysis shows the distribution of orders across product categories.
- State-wise order analysis provides a geographical view of order distribution.
- Warehouse performance comparison helps identify differences in operational activity.

7.2 Executive Overview

The Executive Overview Dashboard provides a summarized view of the overall supply chain business performance. It presents important KPIs and visualizations to help understand revenue, orders, customers, inventory value, and warehouse performance.

- Monthly revenue analysis helps track changes in revenue throughout the year.
- Warehouse-wise order analysis compares the order volumes handled by different warehouses.
- Carrier-wise shipping cost analysis compares transportation expenditure across carriers.
- Warehouse-wise revenue analysis identifies the revenue contribution of individual warehouses.
- Interactive Carrier and Warehouse slicers allow users to filter the dashboard and analyze specific business operations.
- KPI cards provide a quick overview of the main business indicators.

7.3 Warehouse Performance KPIs & Measures

The following KPIs were created for warehouse efficiency and executive-level performance analysis:

KPI / Measure	Purpose
Total Warehouses	Shows the total number of warehouses.
Total Orders	Shows the total number of orders handled.
Total Shipped Quantity	Measures the total quantity of products shipped.

Total Warehouse Capacity	Shows the combined capacity of the warehouses.
On-Time Shipments	Counts shipments delivered within the defined on-time period.
On-Time Rate	Measures the proportion of on-time shipments.
Total Revenue	Shows the overall revenue generated.
Total Customers	Shows the total number of customers.
Inventory Value	Measures the total value of inventory.

7.4 Optimization Analytics

The Optimization Analytics Dashboard is intended to support the identification of operational improvements using supply chain data.

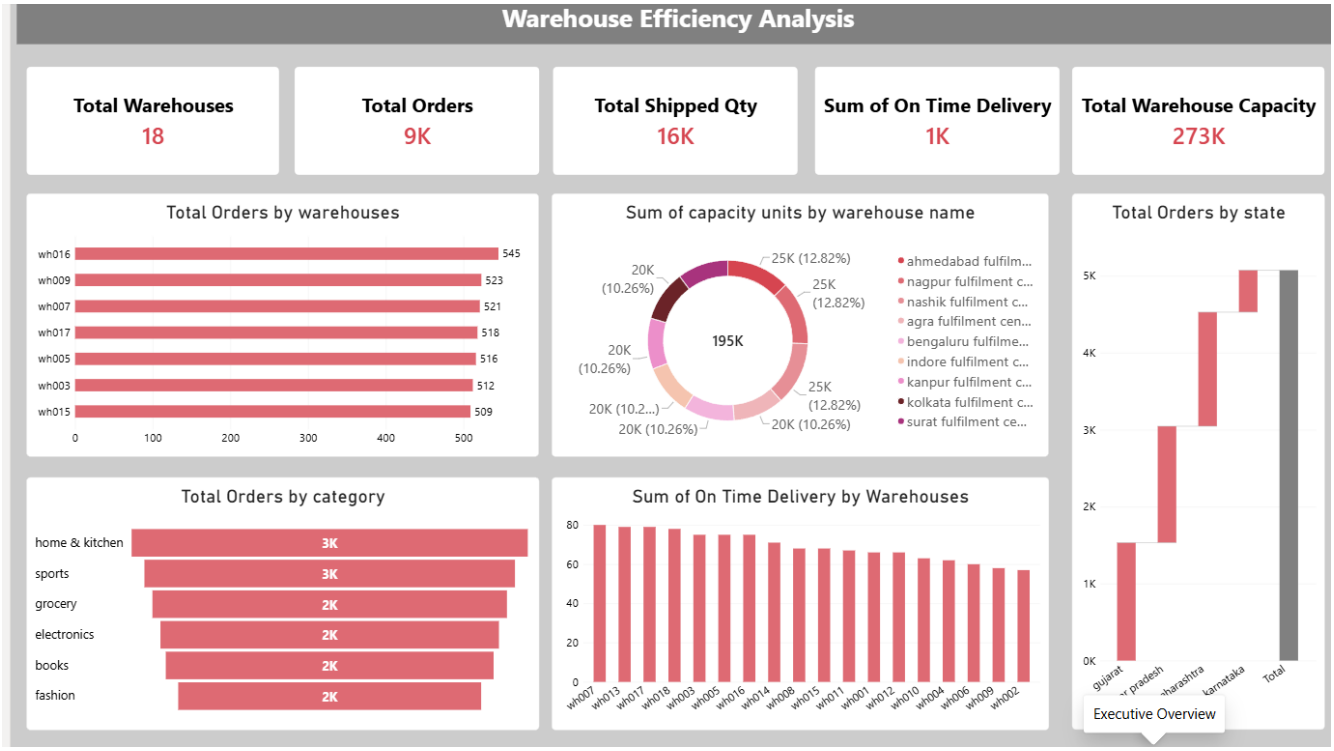
- Warehouse performance comparison helps examine differences in order volumes, capacity, and delivery performance.
- Shipment analysis helps identify transportation and delivery patterns.
- Operational KPI analysis provides information that can support decisions about warehouse and transportation activities.
- Visual comparisons help identify areas that may require further investigation and optimization.

7.5 Milestone 4 Dashboard Screenshots

The dashboard contains KPI cards for

- total warehouses
- total orders
- total shipped quantity
- on-time deliveries
- total warehouse capacity

. It also includes warehouse-wise order analysis, capacity comparison, state-wise orders, category-wise orders, and on-time delivery by warehouse.



The dashboard contains KPI cards for total revenue, total orders, total customers, inventory value, and on-time rate. It also includes monthly revenue trends, warehouse-wise order analysis, carrier-wise shipping costs, and warehouse-wise revenue contribution.



The dashboard presents optimization-related analysis using supply chain performance indicators and visual comparisons.

warehouse_name	Total Shipped Qty	Total orders_items	Total Warehouse Capacity	On Time Shipments	On Time Rate	Total Revenue	Aver
agra fulfilment center	1983	482	20000	68	0.84	85982852.29	
ahmedabad fulfilment center	1404	494	25000	75	0.78	52198318.91	
bengaluru fulfilment center	539	514	20000	75	0.82	47327702.57	
chennai fulfilment center	604	480	5000	78	0.85	30325846.85	
delhi fulfilment center	395	474	8000	67	0.79	54376695.88	
hyderabad fulfilment center	461	444	12000	63	0.84	30036336.92	
indore fulfilment center	966	443	20000	57	0.89	31422212.31	
jaipur fulfilment center	998	459	5000	60	0.82	55983558.93	
kanpur fulfilment center	819	476	20000	62	0.78	38870312.80	
kolkata fulfilment center	495	460	20000	66	0.83	39983606.30	
lucknow fulfilment center	465	486	15000	75	0.82	30394369.69	
ludhiana fulfilment center	513	498	10000	80	0.86	39676729.25	
mumbai fulfilment center	2128	469	5000	79	0.87	37828272.36	
nagpur fulfilment center	505	447	25000	66	0.80	41969770.22	
nashik fulfilment center	1096	497	25000	58	0.81	24980351.47	
patna fulfilment center	938	478	10000	71	0.85	42824744.56	
surat fulfilment center	926	480	20000	68	0.80	31277843.56	
vadodara fulfilment center	517	493	8000	79	0.83	58796744.53	
Total	**15752**	**8574**	**273000**	**1247**	**0.83**	**774256269.40**	

8. Conclusion

The internship project, “Supply Chain Visibility System with Optimization Analytics,” provided practical experience in data analysis, data cleaning, data modelling, DAX calculations, and data visualization using Microsoft Power BI.

The project was completed in four milestones:

- Milestone 1 – Data Modelling & KPI Foundation: Focused on understanding the dataset, cleaning data, creating relationships, developing KPIs, and building the initial dashboard.
- Milestone 2 – Inventory & Delivery Analytics: Focused on inventory quantities, inventory value, inventory turnover, shipment status, shipping costs, and delivery performance.
- Milestone 3 – Supplier & Transportation Analytics: Focused on supplier reliability, supplier-wise shipment analysis, transportation costs, carrier performance, and on-time delivery.
- Milestone 4 – Warehouse Efficiency, Executive Overview & Optimization Analytics: Focused on warehouse performance, capacity, order distribution, revenue analysis, and overall supply chain insights.

Through this project, practical knowledge of Power BI, Power Query, DAX, data modelling, and interactive dashboard development was gained. The dashboards help present complex supply chain information in a structured and understandable format, making it easier to compare performance, identify areas for further analysis, and support data-driven decision-making.

Overall, this internship helped develop technical skills and a better understanding of how data analytics and visualization can be applied to real-world business operations.

9. References

1. Infosys Springboard – Internship learning resources and project guidance. Website: https://infyspringboard.onwingspan.com/

2. Microsoft Power BI – Documentation and learning resources for dashboard development, data visualization, and reporting. Website: https://learn.microsoft.com/en-us/power-bi/

3. Microsoft Power Query – Data cleaning, transformation, and preprocessing. Website: https://learn.microsoft.com/en-us/power-query/

4. Microsoft DAX – Documentation for DAX functions, measures, and calculations used in Power BI. Website: https://learn.microsoft.com/en-us/dax/

5. Microsoft Excel – Spreadsheet tools used for data inspection and preparation. Website: https://support.microsoft.com/excel

6. Supply Chain Management Dataset – Dataset used for the internship project, including customer, product, order, supplier, warehouse, and shipment information.